

MASTER SUMMARY / ALL VOLUMES

Research Summary

Consolidated Findings and Recommendations - Low-Visibility Haulage at Bailadila

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Volume	Master Summary
Programme	Safe & Continuous Haulage under Low-Visibility Monsoon Conditions
Site of Application	Bailadila Iron Ore Mines - Kirandul & Bacheli Complexes, Dantewada, Chhattisgarh
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ABSTRACT

This document consolidates the research package into a single readable account. It states the problem in engineering terms, sets out the central physical finding that determines the architecture, summarises the assessment of existing systems and the gaps they leave, presents the recommended architecture and the reasoning behind it, and closes with a phased roadmap, a risk position and a set of prioritised recommendations. It is intended to be read on its own by a reader who will not read the eleven supporting documents, and to serve as the navigation guide for a reader who will.

Prepared as part of a solution-design research package addressing dense-fog and low-visibility haul road operations in large-scale mechanised open cast iron ore mines. Figures relating to NMDC production, statutory provisions and commercial systems are compiled from public sources; site-specific values marked as indicative require validation against NMDC operational records before use in a formal proposal.

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1. The Problem in One Page

NMDC produced **53.15 MT** of iron ore in FY 2025-26, a 21 per cent increase and the first time an Indian producer has crossed 50 MT. The Bailadila sector - Kirandul and Bacheli - contributes roughly **70 per cent** of that. Aligned with the National Steel Policy 2017 target of 300 MT of domestic steelmaking capacity by 2030-31, requiring 430-437 MT of iron ore annually, NMDC targets **over 100 MT by 2030**, with approximately **80 MT from Bailadila**.

For five months a year, from June to October, monsoon orographic cloud sits on the Bailadila ridge. Visibility on hilltop haul roads falls to **3-5 metres**. Dumper operators reduce speed to a crawl or stop. Cycle times inflate, fleet utilisation falls, and haulage is suspended - repeatedly, because advection fog is transient, and each stop-start cycle carries a fixed restart overhead that is routinely underestimated.

Expressed as engineering rather than inconvenience:

At MOR 3-5 m the atmospheric extinction coefficient is **600-1000 per km**. That is below the safe stopping distance of a loaded rear-dump truck on a 1-in-16 descent at any useful speed. Absent a non-optical means of knowing (a) where the vehicle is laterally within the road, (b) the range and closing rate of forward hazards, and (c) where all other traffic is and what it intends, the only available control is administrative suspension. The resulting loss of effective fleet hours is a constraint on monsoon production today - and becomes a **binding constraint** on the path from ~37 MTPA to ~80 MTPA, because at 80 MTPA there is no spare dry-season capacity to absorb the catch-up.

That reframing is the package's first contribution. This is not primarily a safety problem to be managed by stopping. It is a **capacity recovery problem** whose value scales linearly with the production ramp.

2. Package Structure and How to Read It

Volume	Document	ID	What it does
I - Problem Research	Problem Background	PR-01	Context, meteorology, physics of the operating point, loss model
	Existing Challenges	PR-02	Why every current control fails; constraint register C-01 to C-26; failure modes F1 to F12
	Field Requirements	PR-03	Numbered, testable specification; performance derived from stopping distance
II - Existing Solutions	Existing Systems	ES-01	Survey of mining CAS, AHS, adjacent industries, research
	Technology Comparison	ES-02	Weighted evaluation by layer; recommended technology set; explicit rejections
	Gap Analysis	ES-03	Gaps G1-G6, the architectural inversion, the recommended architecture, roadmap, risks
III - Technology	Radar Research	TC-01	mmWave radar in depth; the residual-sensing layer
	Sensor Research	TC-02	Positioning, road model, thermal, LiDAR, visibility, fusion
	AI/ML Research	TC-03	Where learning belongs and where it does not
	Communication Technology	TC-04	Message set, bearer evaluation, propagation, security
IV - References	Research Papers	RF-01	Annotated bibliography with evidence-class ratings

Volume	Document	ID	What it does
	Government Reports	RF-02	Statutory and policy instruments; compliance matrix
	Standards	RF-03	Standards families; procurement checklist

Suggested reading paths. A decision-maker: this summary, then ES-03. An engineer specifying the system: PR-03, then ES-02, then the Volume III document for their layer. A procurement officer: PR-03, then RF-03 Section 13. A regulator or safety authority: PR-01, PR-02, then RF-02.

3. The Central Physical Finding

Everything in the package follows from one table.

Fog attenuation is governed by Mie scattering, which peaks when droplet size is comparable to wavelength. Fog droplets are 1-20 μm .

Modality	Wavelength	Size parameter (10 μm droplet)	Behaviour at MOR 3-5 m
Human eye, visible camera	0.55 μm	~ 57	Blind
NIR LiDAR	905 nm	~ 35	Range collapse
SWIR LiDAR	1550 nm	~ 20	Degraded
LWIR thermal	10 μm	~ 3	Partial penetration only
mmWave radar	3.9 mm	~ 0.008	Essentially unaffected
Cooperative RF	0.1-1 m	$\sim 1\text{e-}5$	Unaffected
GNSS L-band	~ 0.2 m	$\sim 1\text{e-}5$	Unaffected

Supporting measurements: at 77 GHz, attenuation in 25 mm/hr heavy rain is approximately **10 dB/km** against approximately **35 dB/km** for 905 nm LiDAR under comparable conditions. Fusion studies report **LiDAR alone failing beyond approximately 70 m in dense fog while a radar-fused system sustains detection beyond 100 m**.

The link-budget consequence is stark. Over a 120 m two-way path, heavy fog costs radar roughly **one decibel**. It costs an optical sensor everything, many times over, within the first few metres.

Conclusion, stated as a design rule. No amount of processing recovers information that never reached the sensor. Dehazing, contrast enhancement and generative reconstruction cannot invert a signal extinguished below the noise floor. Any architecture whose *primary* forward-sensing layer is optical is disqualified. The primary layers must be millimetre-wave radar and cooperative RF, with optical modalities contributing only in the visibility classes where they remain valid.

4. What Exists and What It Leaves Undone

Every constituent capability already exists in commercial or proven form.

Capability	Where it exists today
Fog-tolerant long-range sensing	Automotive 77/79 GHz radar - a mass-market commodity
Sub-metre positioning through GNSS shadowing	The autonomous haulage navigation stack, in production use
Maintained digital haul road models at production scale	Autonomous haulage operations

Capability	Where it exists today
Cooperative broadcast and conflict computation	Mining CAS (65,000+ units, 100+ sites), aviation ADS-B, maritime AIS
Synthetic terrain display from a database	Aviation Synthetic Vision Systems
Graded operating states triggered by measured visibility	Aviation low-visibility procedures; highway fog warning systems
Central movement authority in zero visibility	Rail ERTMS/ETCS

No existing system combines them for this operating point. Mining CAS gives cooperative awareness without lane guidance. Autonomous haulage gives lane guidance at inapplicable economics, on gentler terrain, with an **unproven dense-fog envelope** - AHS may face the same problem with the driver removed. Visibility systems give graded modes without vehicle guidance.

The six gaps:

Gap	Statement
G1	No mining system tells a manned truck operator, in dense fog, where the edge of the road is. The largest gap and the one whose closure unlocks most of the benefit.
G2	Deployed proximity detection operates at 10-15 m. The stopping-distance analysis requires 50-80 m.
G3	Fog response is subjective, mine-wide and binary. Bailadila fog is stratified by elevation and patchy.
G4	No integrated picture combining vehicles, visibility, road state and production consequence.
G5	No vendor publishes detection performance against calibrated MOR. A market-wide evidence gap.
G6	Capable systems are OEM-native or autonomy-class; neither fits a mixed brownfield retrofit.

Why the gaps persist. Products were specified against the pedestrian blind-zone hazard, correctly, and inherited those specifications. The global market is shaped by dust, not monsoon fog. Demonstrations happen in clear weather, where cameras look excellent. Investment went into removing drivers rather than assisting them. **These gaps will not close on their own before 2030.**

5. The Architectural Inversion

The problem as usually framed - *how do we let the operator see through fog?* - is close to unsolvable at MOR 3-5 m.

Aviation asks a different question: *how do we reduce what the operator needs to see?* That question has a complete answer.

- ? The **static world** - where the road is, its curve, gradient, berm, junction, tipping edge - never needs to be seen. It is known in advance. It needs only to be *located*. That is a precise position against an accurate map. **This is Synthetic Vision.**
- ? The **cooperative world** - every other truck, dozer, grader, tanker, light vehicle, equipped person - never needs to be seen either. Each knows where it is and can say so over a link fog does not touch. **This is ADS-B, AIS and mining CAS.**
- ? Only the **residual world** - rockfall, washout, a stalled unequipped vehicle, an untagged person, an animal - genuinely requires sensing. That is a much smaller problem, and 77/79 GHz radar solves it.

The static world comes from the map. The cooperative world comes from the radio. Only the residual needs a sensor. Framed this way, dense fog reduces a complete perception problem to a bounded detection problem - and the bounded problem has a commodity solution.

Two corollaries, both load-bearing:

The map becomes safety-critical. A stale road model in dense fog will confidently draw a road that is not there. Version control, validity timestamps, field update within a shift, divergence monitoring against driven paths, and visibly degraded rendering of stale sections are not refinements. They are what makes the architecture safe.

The radar layer must never be economised away. It covers exactly what the map and radio cannot know about - and those hazards become *more* common during monsoon, not less.

6. The Recommended System

Fog-Independent Haulage Guidance and Awareness System (FIHGAS).

Eight design principles: fog-independence by physics not processing; know rather than see; two independent primary layers; on-vehicle autonomy for all safety processing; graded instrumented response; honest degradation; advisory first with intervention architecturally separated; retrofit and vendor-neutral.

Layer	Content
L0 Environment	Forward-scatter visibility meters by elevation zone, reference transmissometer, met sensing, nowcasting
L1 Position	RTK GNSS, dual-antenna heading, tactical IMU tightly coupled, odometry, radar-odometry aiding, protection level
L2 Map	Digital haul road model: centreline, edges, lanes, gradient, curvature, zones, tipping edges, with validity
L3 Cooperative	V2V broadcast at 10 Hz - identity, class, position, confidence , velocity, heading, state, system health
L4 Residual sensing	4D imaging radar forward and quarters; short-range radar 360 deg; magnetic PDS for personnel; thermal and camera secondary
L5 Fusion and threat	Track fusion, prediction against the road model , time-to-conflict, graded alerting, self-degradation detection
L6 Interface	Plan view primary; synthetic perspective promoted at V3-V4; directional audio; haptic; persistent health and effective range
L7 Coordination	Common operating picture with visibility overlay, movement authority, re-dispatch, event log, FMS integration
L8 Analytics	Nowcast training, KPI reporting, alert tuning, optional digital twin

Graded operating classes, replacing the present binary go/no-go:

Class	MOR	Posture
V0/V1	>50 m	Normal; system in background
V2	20-50 m	Assisted - advisory speed, forward alerting
V3	5-20 m	Guided - lane guidance primary reference, slot-managed critical sections
V4	<5 m	Minimum-speed guided movement on instrumented routes, or controlled suspension

The production recovery lies in converting V2 and V3 from suspension triggers into productive classes.

Honest degradation is what separates a trusted system from a disabled one. GNSS outage beyond coasting limit withdraws lane guidance *visibly*; a blocked radar drops the displayed effective range to zero; a stale road section renders in a distinct degraded style. **The system never presents a normal-looking picture it cannot support.**

7. Performance Requirements and Where They Come From

Requirements are derived, not asserted. Stopping distance on a wet 1-in-16 descent, with a 1.5 s perception-reaction time and an indicative effective deceleration coefficient of 0.25:

Speed	Stopping distance
10 km/h	8.3 m
20 km/h	24.8 m
30 km/h	49.7 m
40 km/h	82.8 m

Two conclusions follow immediately. **At MOR 3-5 m, visual operation cannot exceed about 5 km/h** - which is exactly what is observed. And **a 10-15 m proximity envelope supports about 12-15 km/h at best**, which invalidates the deployed generation of mining proximity devices for this application.

Requirement	Value	Derivation
Detection, vehicle target	80 m minimum, 120 m goal	30 km/h stopping distance x 1.5 margin
Detection, person	25 m, plus full 360 deg blind zone	Consequence-weighted
Horizontal position accuracy	<0.30 m (95%)	Lane margin is 1-2 m
Heading accuracy	<0.3 deg, dual-antenna	1 deg = 1.4 m lateral error at 80 m
Position continuity	No gap >2 s	Guidance cannot flicker
End-to-end alert latency	<200 ms	Budgeted across sensing, fusion, decision, rendering
V2V latency / rate	<100 ms / >=10 Hz	Closing-rate computation
False alert rate	<1 per vehicle-hour	The most important human-factors requirement
System availability	>=99%	Silent unavailability is worse than none

Detection performance must be **demonstrated against calibrated MOR** spanning V2, V3 and V4, with an independent visibility instrument co-located. No standard covers this; the programme must write the test specification.

8. The Human Factors Position

The package's assessment, consistent across the sector record: **systems of this class fail operationally far more often than they fail technically**, and the two highest-rated risks in the register are both human.

? **R3 - Nuisance alerts.** A distance-envelope system alerts constantly in queues, passing manoeuvres and spotting - the situations that occur most often. Operators disengage. Mitigation is trajectory-aware alerting on time-to-conflict, context suppression bounded by a floor below which the system always warns,

single-action operator nuisance reporting, and treating the measured false-alert rate as a headline KPI from week one.

- ? **R7 - Partial fitment.** An uninstrumented machine is invisible to the cooperative layer, and an operator who trusts the display may not account for it. Mitigation: the radar layer is mandatory precisely so that unequipped machines remain detectable, and the interface must distinguish cooperative from sensed contacts.
- ? **R5 - Over-trust.** A fluent, confident interface communicates competence independently of actual performance. In a system used by operators who cannot verify its output against what they see - because they cannot see - that is a hazard. Mitigation: persistent display of effective sensing range and system health.
- ? **R6 - Industrial relations.** Systems introduced as safety tools with agreed, transparent data-use policies are accepted. Systems perceived as disciplinary instruments are resisted or defeated. **The data governance framework must be agreed before any vehicle is fitted.** It costs nothing but attention and cannot be retrofitted after trust is lost.

9. Regulatory Position

The regulatory direction already favours this class of system, which is a significant advantage.

- ? **DGMS Technical Circular 03 of 2024** already mandates Proximity Detection Systems and rear-view cameras on all opencast dumpers, daily pre-operational checklists, certified operators, and a 1-in-16 gradient limit.
- ? **DGMS Technical Circular 06 of 2020** already requires proximity warning devices detecting static and moving objects, with warnings **adjusted by speed and direction** - which is the trajectory-aware principle, already in regulation, though commonly implemented only as a short-range envelope.
- ? **MMR 1961** and the Mines Act 1952 govern; responsibility attaches to lease holder and contractor alike.

The proposed system therefore **extends an existing obligation rather than introducing a new category** - materially easier to justify internally and to discuss with DGMS.

The one structural regulatory issue: any function taking control authority - speed enforcement, automatic braking, autonomy - requires an approval pathway with limited Indian precedent. The package's response is principle P7: **advisory and intervention must be architecturally and contractually separated**, so that Phases 1-3, carrying the majority of the benefit, are never gated by Phase 4 approval.

10. Phased Roadmap

Phase	Scope	Value
0 - Baseline	Visibility sensor network. Extract historical fog stoppage hours, stop-start counts and fog cycle times from existing FMS and shift records. GNSS availability survey. RF propagation survey. Instrumented radar campaign through a monsoon. DGMS engagement. Spectrum confirmation.	The business case, the acceptance baseline, the nowcast training data, and validation of the central architectural assumption. Lowest cost in the programme; prerequisite for everything.
1 - Awareness	Fleet-wide cooperative V2V plus personal devices. RTK infrastructure. In-cab plan view. Control room picture with visibility overlay. FMS integration.	Visibility-independent fleet awareness. V2 becomes defensible. Exceeds existing DGMS PDS mandates.
2 - Guidance	GNSS/INS on haul trucks. Road model creation and field-update workflow. Synthetic perspective view. 4D imaging radar. Full fusion and graded alerting. Visibility-coupled advisory speed.	The core capability. V3 guided operation becomes defensible. Most of the production recovery.

Phase	Scope	Value
3 - Optimisation	Nowcasting. Slot management. Thermal layer. Vehicle-derived visibility. Alert tuning on measured data. Digital twin.	Fewer stop-start events, faster restart - anticipatory rather than reactive.
4 - Intervention	Speed enforcement or automatic braking, DGMS-approval gated. Optional selective autonomy.	Residual risk reduction. Deliberately last; must never gate Phases 1-3.

11. The Single Highest-Value Action

Three separate Phase 0 activities each require one vehicle driving the full haul road network over a period spanning a monsoon, with logging instrumentation:

- ? The **GNSS availability and accuracy survey** - determining where RTK holds, where it degrades, and therefore how much inertial coasting capability is needed.
- ? The **RF propagation survey** - determining whether 5.9 GHz coverage is adequate across switchbacks and pit bottoms, or whether a sub-GHz bearer is required. **This decides the V2V bearer selection.**
- ? The **instrumented radar campaign** - a candidate radar and a calibrated visibility meter logging continuously, producing the first MOR-referenced detection performance data for a mining haul road anywhere.

Combine them into a single Phase 0 field campaign. One vehicle, one instrumentation fit, one dataset, one monsoon. This produces the positioning, communications and radar evidence base simultaneously, at a fraction of the cost of three exercises, and it resolves the three largest technical uncertainties in the programme before a single procurement decision is made.

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It should be initiated at the **earliest available monsoon window**. Fog-condition data can only be collected during a limited annual window; a missed window costs a full year (risk R10). This is the highest-value action available to the programme and it is inexpensive.

12. Risk Position

The fourteen-entry register in ES-03 Section 8 resolves to four positions worth stating at summary level.

- ? **The technical risk is concentrated and testable.** R1 (radar performance at MOR 3-5 m on site) and R2 (GNSS availability in pit geometry) are the two assumptions the architecture rests on. Both are resolved by the Phase 0 campaign, cheaply and early. This is an unusually favourable risk structure.
- ? **The operational risk is higher than the technical risk.** R3 (nuisance alerts) and R7 (partial fitment) are rated highest in combined likelihood and impact, and both are human-factors risks with well-understood mitigations that must be designed in rather than added later.
- ? **The schedule risk is seasonal and unforgiving.** Every fog-condition validation activity is monsoon-gated. **The programme calendar must be planned backwards from monsoon windows**, not forwards from a start date.
- ? **The benefit risk has an open question.** Reporting identifies the single-track Kirandul-Kothavalasa rail line as a historical dispatch ceiling on the landlocked Bailadila sector, subject to monsoon landslides through the Ananthagiri Ghats. **If rail evacuation is also constrained during the same months, recovered haulage hours may be partly absorbed by inability to despatch.** Phase 0 must establish whether the binding

monsoon constraint is mine-side haulage, rail evacuation, or both. This materially affects the realisable benefit and should be resolved before large commitment.

13. What This Package Does Not Establish

Stated plainly, because the package's credibility rests on the distinction.

- ? **It does not predict radar detection range at Bailadila.** The quantitative fog results relied upon - notably the finding that LiDAR fails beyond ~70 m while fused systems sustain beyond 100 m - are **simulation-derived**. They establish which modality is correct by a wide margin. They do not predict performance in monsoon orographic cloud on an ore ridge.
- ? **It does not quantify the production loss.** The loss model in PR-01 Section 4.3 is a structure with four levers, not a calculation. Populating it requires NMDC's own FMS and shift records.
- ? **It does not size the benefit.** The benefit structure in ES-03 Section 9 lists eight streams and the inputs needed. It asserts no value.
- ? **It relies on indicative meteorological figures.** The rainfall and humidity bands in PR-01 are regional approximations framing the problem. IMD station data and NMDC site observations must replace them.
- ? **It adopts indicative braking parameters.** The stopping-distance derivation uses a conservative placeholder deceleration coefficient. **Measured braking performance for the actual fleet must replace it** - if real deceleration on wet ore fines is materially lower, the required detection range rises.
- ? **No vendor's dense-fog performance is verified.** The industry publishes none. That is gap G5, and it is why acceptance testing against measured MOR is non-negotiable.

The annotated bibliography (RF-01) carries an evidence-class rating on every entry and an explicit evidence-strength table for the package's principal claims, so that a reader can see exactly which conclusions rest on physics, which on industry disclosure, and which on nothing yet.

14. Prioritised Recommendations

Immediate, independent of any procurement decision:

- ? **Launch the combined Phase 0 field campaign** at the earliest monsoon window - GNSS survey, RF survey and instrumented radar logging, one vehicle, one dataset.
- ? **Deploy the visibility sensor network now.** It is the cheapest element of the architecture, has no dependencies, immediately converts the fog decision from subjective to instrumented, and begins accumulating nowcast training data that cannot be recovered later.
- ? **Extract the historical loss baseline** from existing FMS and shift records. It costs analyst time and produces the business case.
- ? **Establish whether rail evacuation is co-constraining during monsoon months.** This determines realisable benefit.
- ? **Open the DGMS dialogue**, presenting the programme as an extension of existing Circular 03/2024 and 06/2020 obligations.
- ? **Confirm the spectrum position** for 76-81 GHz radar and for the candidate V2V bearers. Long lead time; risk R9.
- ? **Agree the data governance framework** before any vehicle is fitted.

In specification and procurement:

- ? **Specify against "know rather than see."** Reject any proposal whose primary forward-sensing layer is optical.

- ? **Make MOR-referenced acceptance testing contractual.** Without it, NMDC will procure clear-weather performance for a fog problem.
- ? **Make the measured false alert rate a headline acceptance criterion,** with a named owner and a tuning budget, measured from week one.
- ? **Require an open, documented road model format** maintainable by NMDC's survey department without the supplier. Otherwise field update within a shift is unachievable in practice.
- ? **Separate advisory from intervention** architecturally and contractually.
- ? **Require complete cooperative-layer fleet coverage before relying on it,** and retain the radar layer as mandatory so partial coverage never creates false confidence.
- ? **Replace every indicative parameter in this package** - braking performance, rainfall climatology, fleet productivity - with measured site values before the specification is issued.

Strategically:

- ? **Treat the resulting MOR-referenced performance dataset as an asset.** It would be the first of its kind in mining, is directly transferable to other Indian monsoon-affected operations, is likely to be of direct interest to DGMS, and materially strengthens NMDC's position in specifying and procuring future systems.
- ? **Evaluate the programme not only on tonnes recovered today but on whether the 2030 Bailadila target is achievable without it.** At 37 MTPA, monsoon shortfall is recoverable in the dry season. At 80 MTPA, against a fleet and haul road network near their geometric limits, it is not.